Customer Sales Analysis

Week 5 Project Report: Advanced Data Manipulation with Pandas

Executive Summary

This report analyses 100 sales orders (1 January to 9 April 2024) together with a customer file of 500 customers. The sales data totals **Rs.12,365,048**, with an average order of **Rs.123,650**. The main findings are:

- **Laptops lead:** Rs.3,889,210 (31.5% of sales). Laptop, Tablet and Phone together bring 78% of all sales.
- **North and South are the strongest regions** (32.2% and 30.2% of sales). West has almost as many orders as South but much smaller orders (average Rs.81,689).
- **Customers leave early:** 63.1% of customers in their first 12 months left, compared with 0.0% of customers who stayed longer than 12 months.
- **Contract type matters:** month-to-month customers churn at 20.6%, against 4.3% for one-year contracts.
- **Customers who left spent less:** their average order was Rs.53,163 compared with Rs.131,482 for customers who stayed.
- **A few large orders matter a lot:** the top 10 orders make 27.1% of all sales.

Top recommendations: (1) build an onboarding and early-engagement programme for first-year customers, (2) offer incentives to move month-to-month customers to one-year contracts, (3) protect the high-value accounts, (4) focus Laptop promotion in the North and South, and (5) collect multi-item order data to enable cross-selling analysis.

1. Data and Method

Files used: sales_data.csv (100 orders, 7 columns: Date, Product, Quantity, Price, Customer_ID, Region, Total_Sales) and customer_data.csv, downloaded as customer_churn.csv (500 customers, 9 columns: CustomerID, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen, Churn). In the Churn column, 1 means the customer left and 0 means the customer stayed.

Method (pandas):

- **Explore:** checked rows, columns, data types, missing values and duplicates. Both files had no missing values and no duplicate rows.
- **Clean:** removed extra spaces and standardised text with string methods (strip, title); converted Date to a real date and extracted year, month, month name and quarter; created an order-size column (Small, Medium, Large).
- **Join:** merged the sales data with the customer data (left merge, many-to-one) so that each order carries its customer's contract, tenure and churn status.
- **Analyse:** used groupby aggregations (product, region, month, contract), a top-customer ranking, pivot tables, and churn-rate calculations.
- **Visualise:** created 5 charts with matplotlib (bar, line, pie, heatmap and a pair of churn bar charts).

Important assumption about joining the files

The two files do not use the same customer ID format: the customer file uses IDs like C00001 and the sales file uses IDs like CUST001, so **no IDs match exactly**. Both IDs end with a running number, so the

number (1, 2, 3 and so on) was extracted from each ID and used as the join key. This treats CUST001 and C00001 as the same customer. It is an assumption, not a confirmed fact. All 100 orders found a match under this rule, but the customer-level results in this report (contract type, tenure and churn of the buyers) should be confirmed with the data owner before important business decisions.

2. Sales Analysis

2.1 Sales by product

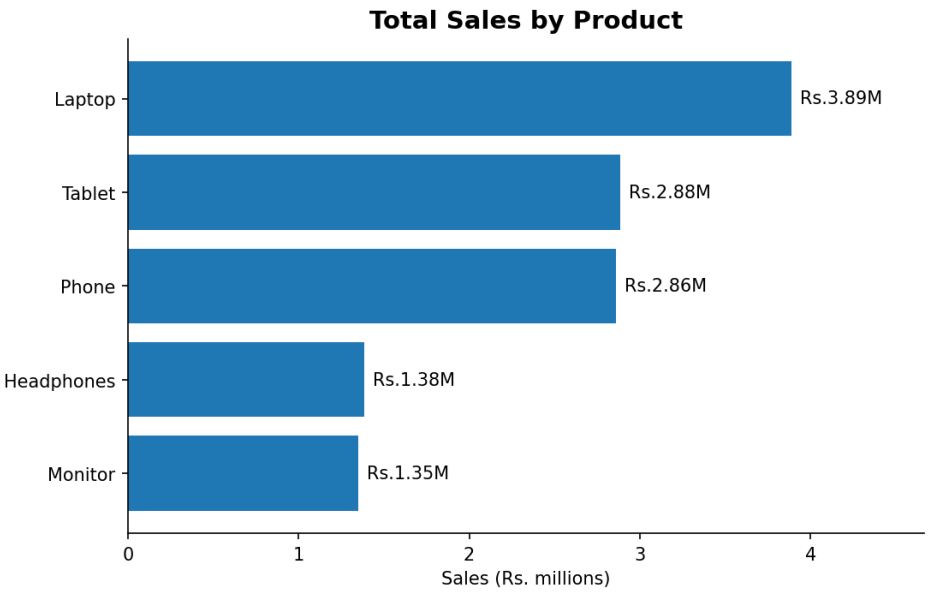


Chart 1: Total sales by product (bar chart)

Product	Total sales	Share	Orders	Units sold	Average order
Laptop	Rs.3,889,210	31.5%	24	136	Rs.162,050
Tablet	Rs.2,884,340	23.3%	26	127	Rs.110,936
Phone	Rs.2,859,394	23.1%	20	101	Rs.142,970
Headphones	Rs.1,384,033	11.2%	15	48	Rs.92,269
Monitor	Rs.1,348,071	10.9%	15	66	Rs.89,871

Laptop is the best-selling product by revenue and by units (136 units) and has the highest average order (Rs.162,050). Headphones and Monitor have the lowest sales (Rs.1,384,033 and Rs.1,348,071).

2.2 Monthly trend

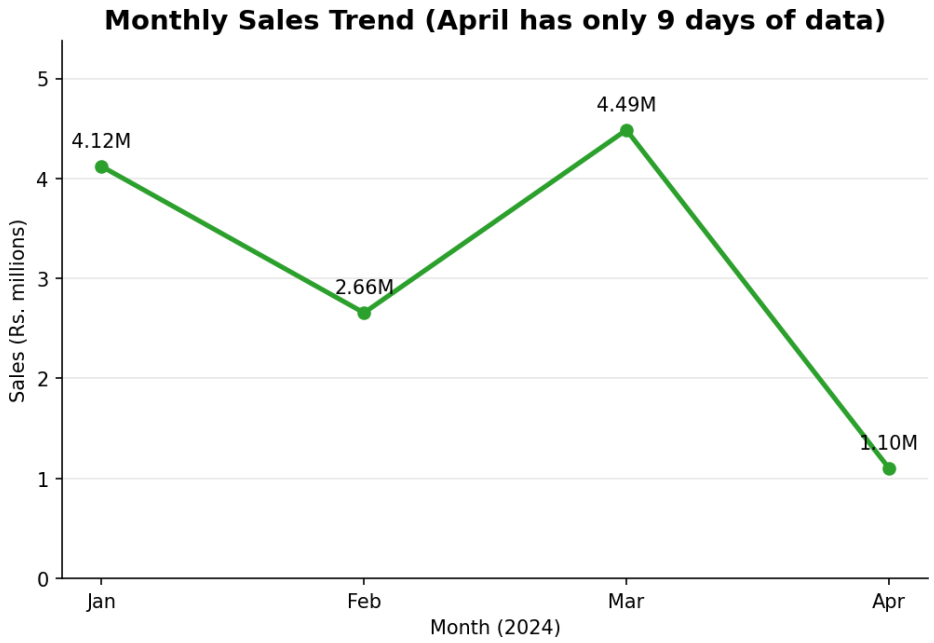


Chart 2: Monthly sales trend (line chart)

Month	Total sales	Orders	Average order
Jan 2024	Rs.4,120,524	31	Rs.132,920
Feb 2024	Rs.2,656,050	29	Rs.91,588
Mar 2024	Rs.4,485,006	31	Rs.144,678
Apr 2024	Rs.1,103,468	9	Rs.122,608

March was the best month (Rs.4,485,006). February was the weakest full month: sales fell 36% from January because orders were smaller (average Rs.91,588), not because there were fewer orders. April looks very low only because the data ends on 9 April (9 orders), so it should not be compared directly with full months. With only about three months of data, real seasonal trends cannot be confirmed.

2.3 Sales by region

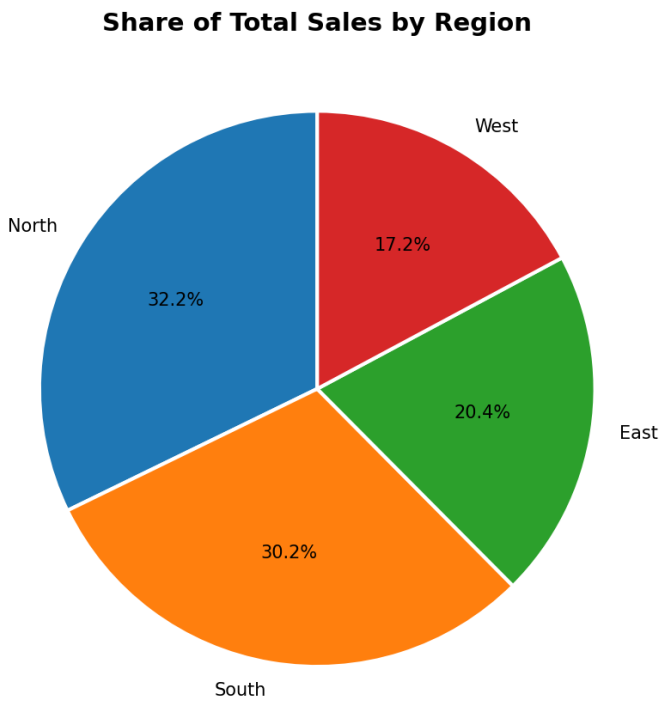


Chart 3: Share of total sales by region (pie chart)

Region	Total sales	Share	Orders	Average order
North	Rs.3,983,635	32.2%	28	Rs.142,273
South	Rs.3,737,852	30.2%	27	Rs.138,439
East	Rs.2,519,639	20.4%	19	Rs.132,613
West	Rs.2,123,922	17.2%	26	Rs.81,689

North and South together bring 62% of sales. West has 26 orders, almost as many as South (27), but its average order is only Rs.81,689, so West is behind because of order size, not because of fewer customers.

2.4 Which products sell where

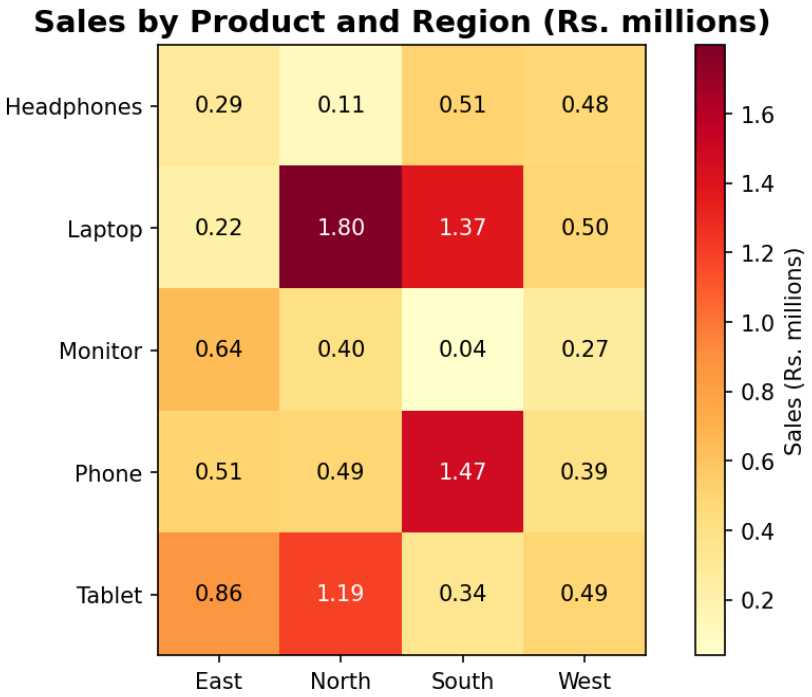


Chart 4: Sales by product and region, from a pivot table (heatmap, Rs. millions)

The strongest combination is Laptop in the North (Rs.1,798,206), followed by Phone in the South (Rs.1,471,428) and Laptop in the South (Rs.1,373,120). Tablets sell well in the North (Rs.1,191,954). The weakest combination is Monitor in the South (Rs.39,924).

3. Customer Analysis

3.1 Top customers

Customer	Product	Region	Order value	Contract	Tenure (months)
CUST016	Phone	South	Rs.373,932	Month-to-month	65
CUST007	Laptop	South	Rs.363,870	Two year	66
CUST083	Laptop	North	Rs.350,888	Two year	27
CUST073	Headphones	West	Rs.349,510	Two year	13
CUST020	Laptop	North	Rs.333,992	Month-to-month	26

Each customer in the sales file placed exactly one order, so customer value here means the value of that order. The top 10 orders make 27.1% of all sales. Two of the top five customers (CUST016 and CUST020) are on month-to-month contracts, so they could leave at short notice.

Sales by contract type

Contract type	Total sales	Orders	Average order
Month-to-month	Rs.3,766,277	35	Rs.107,608
One year	Rs.5,216,183	36	Rs.144,894
Two year	Rs.3,382,588	29	Rs.116,641

One-year contracts bring the most revenue (Rs.5,216,183) and the highest average order (Rs.144,894).

3.2 Customer retention and churn

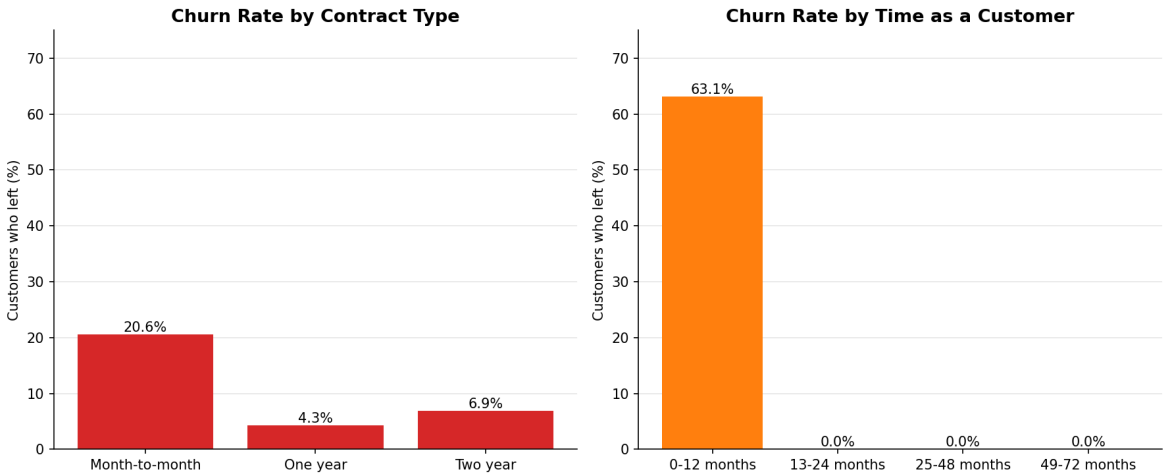


Chart 5: Churn rate by contract type and by time as a customer (bar charts, all 500 customers)

Group	Customers	Churn rate	Retention rate
Month-to-month contract	170	20.6%	79.4%
One year contract	186	4.3%	95.7%
Two year contract	144	6.9%	93.1%
0-12 months	84	63.1%	36.9%
13-24 months	75	0.0%	100.0%
25-48 months	171	0.0%	100.0%
49-72 months	170	0.0%	100.0%
All customers	500	10.6%	89.4%

The overall churn rate is 10.6% (retention 89.4%). Churn is concentrated at the start of the customer relationship: 63.1% of customers with 12 months or less left, while no customer with more than 12 months of tenure left. Month-to-month contracts also have a much higher churn rate than one-year and two-year contracts.

Among the 100 buyers in the sales file, 10 left. Their average order was Rs.53,163, compared with Rs.131,482 for the 90 who stayed. Together the customers who left account for Rs.531,627 (4.3% of sales).

4. Answers to the Analysis Questions

Question	Answer
Who are our most valuable customers?	The largest single orders come from CUST016 (Phone, South), CUST007 (Laptop, South), CUST083 (Laptop, North), CUST073 (Headphones, West) and CUST020 (Laptop, North). The top 10 orders make 27.1% of sales.
What products sell best together?	This cannot be measured with this data: every customer placed one order with one product, so there are no multi-product baskets. The product-by-region heatmap shows where each product sells best instead.
Which regions have the highest sales?	North (Rs.3,983,635) and South (Rs.3,737,852), followed by East and West.
What are the seasonal trends?	March was the strongest month and February the weakest full month. Only about three months of data are available, so seasonality cannot be confirmed.
How can we improve customer retention?	Focus on the first 12 months, when most churn happens, and move month-to-month customers to longer contracts (see recommendations).

5. Business Recommendations

1. **Fix the first year.** Churn is 63.1% in the first 12 months and 0% afterwards. Introduce an onboarding programme, a check-in after the first month, and early offers for new customers.
2. **Move customers to longer contracts.** Month-to-month churn is 20.6% against 4.3% for one-year contracts. Offer a discount or bonus for switching to a one-year plan, starting with high-value month-to-month customers such as CUST016 and CUST020.
3. **Protect high-value accounts.** The top 10 orders make 27.1% of sales. Give these customers a named contact and regular follow-up.
4. **Double down on what works.** Promote Laptops in the North and South and Phones in the South, where Rs.1,798,206, Rs.1,373,120 and Rs.1,471,428 of sales come from.
5. **Lift order value in the West.** The region's average order (Rs.81,689) is far below the other regions. Bundles and upselling could help.
6. **Revive weak product-region pairs.** Monitor sales in the South are only Rs.39,924. Test a targeted promotion there.
7. **Collect better data.** Record multi-item orders so cross-selling can be measured, use one customer ID format in all systems, and gather at least a full year of sales to study seasonality.

6. Limitations

- The sales file has only 100 orders over about three months, and April has only 9 days of data.
- The customer IDs in the two files do not match exactly. The join used the number inside each ID, which is an assumption.
- Each sales customer placed only one order, so repeat purchases, lifetime value and cross-selling cannot be measured.
- There is no cost or profit data, so the analysis covers sales only.
- Churn results come from a customer file that may not describe the same customers as the sales file.

7. Conclusion

Laptops, the North and South regions, and one-year contracts drive the most value. The biggest risk is customers leaving in their first year, especially on month-to-month contracts. Acting on the recommendations above can protect revenue and improve retention. The analysis follows the full pandas workflow (load, clean, join, group, pivot and visualise), and the join assumption should be verified before the results are used for business decisions.